

Analytic Geometry

Slope · Graphing Lines · Parallel & Perpendicular Lines · Length of a Segment · Midpoint

MTH1W · Homework — Independent Practice

Name: _____

Date: _____

Goal: Practise slope, graphing, parallel/perpendicular lines, length and midpoint independently.

Show your full working in the space provided under each question. Levels: L1 fluency, L2 application, L3 thinking. Full worked solutions are in the Answer Key.

Level 1 — Fluency

1. Find the slope of the line $y = 6x + 1$. **[Slope]**
2. State the slope and the y-intercept of $y = x - 5$. **[Graphing]**
3. State the slope of any line parallel to $y = 5x - 3$. **[Parallel]**
4. State the slope of a line perpendicular to a line with slope 4. **[Perpendicular]**

5. Find the length of the segment from $(0, 0)$ to $(3, 4)$. **[Length]**

6. Find the midpoint of $(2, 2)$ and $(8, 6)$. **[Midpoint]**

Level 2 — Application

7. Find the slope of the line through $(2, -1)$ and $(6, 7)$. **[Slope]**

8. For $y = -2x + 8$, find the x-intercept and the y-intercept. **[Graphing]**

9. Write the equation of the line parallel to $y = 4x - 7$ that passes through $(0, 3)$.
[Parallel]

10. State the slope of a line perpendicular to $y = (2/3)x + 1$. **[Perpendicular]**

11. Find the length of the segment from $(-1, 2)$ to $(11, 7)$. **[Length]**

12. Find the midpoint of $(-3, 5)$ and $(7, -1)$. **[Midpoint]**

Level 3 — Thinking

13. The line through $(1, 4)$ and $(5, k)$ has slope 3. Find the value of k . **[Slope]**
14. Rewrite $4x - 2y = 8$ in the form $y = mx + b$, then state the slope and y-intercept.
[Graphing]
15. Determine whether the line through $(0, 1)$ and $(3, 10)$ is parallel to $y = 3x - 4$.
[Parallel]

16. Are $y = 3x + 2$ and $y = -(1/3)x - 5$ perpendicular? Justify. **[Perpendicular]**
17. A triangle has vertices $A(1, 1)$, $B(1, 5)$ and $C(4, 1)$. Find its perimeter. **[Length]**
18. $M(3, -2)$ is the midpoint of $P(-1, 4)$ and Q . Find the coordinates of Q . **[Midpoint]**